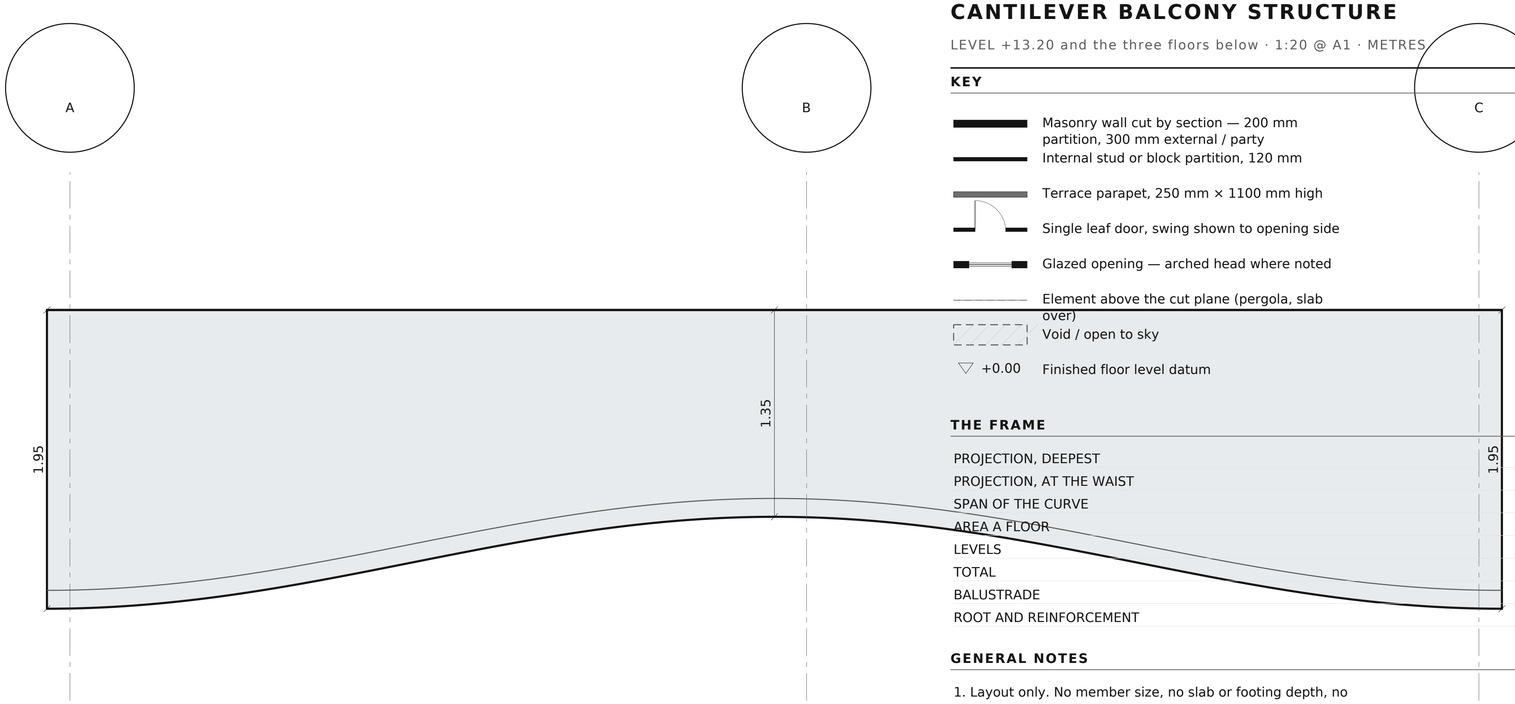
CANTILEVER BALCONY STRUCTURE

LEVEL +13.20 and the three floors below · 1:20 @ A1 · METRES



SLAB EDGE — THICKNESS, ROOT DEPTH AND REINFORCEMENT BY THE ENGINEER

KEY

- Masonry wall cut by section — 200 mm partition, 300 mm external / party
- Internal stud or block partition, 120 mm
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- +0.00 Finished floor level datum

THE FRAME

PROJECTION, DEEPEST	1.95 m
PROJECTION, AT THE WAIST	1.35 m
SPAN OF THE CURVE	9.50 m
AREA A FLOOR	15.7 m²
LEVELS	4
TOTAL	62.7 m²
BALUSTRADE	glass, 120 mm inboard
ROOT AND REINFORCEMENT	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. A 1.95 m cantilever on a curve, repeated on four levels, is the single most engineering-sensitive element in this building. It is drawn at 1:20 for that reason and for no other.

4. The curve, the span, the projection at every point and the balustrade line are exact. The slab thickness, the root depth into the floor behind, the top steel and the deflection and vibration checks are entirely the engineer's.

5. Grid line 1 runs along the face the cantilever springs from. The balcony projects south of it and is unsupported at its far edge on all four levels.

FIVE-STOREY VACATION VILLA — SHAMANDEEL, QUESNA, MENOUFIA, EGYPT

DRAWING

CANTILEVER BALCONY STRUCTURE

DRAWING No	REVISION	SCALE	DATE
STR-500	**P19**	**1:20 @ A1**	**2026-10-07**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction. Do not scale; work to figured dimensions only.

REVISIONS

P16 2026-10-07 neighbours as surveyed: west 3 storeys and blind, east single storey
P17 2026-10-07 ARC-000 deliverables list; title corrected to five storeys
P18 2026-10-07 scheme A adopted: serpentine balcony on every plan; site named
P19 2026-10-07 serpentine waist deepened 0.75 to 1.35 m; 15.7 m² of balcony a floor